

# A Novel Set-Valued Sensor Fault Diagnosis Method for Lithium-Ion Battery Packs in Electric Vehicles

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**Abstract**—Sensor fault diagnosis is of great significance to ensure safe battery operation. This paper proposes a novel sensor fault diagnosis method that achieves the simultaneous fault detection, fault source and type identification, and fault estimation in a comprehensive way. Specifically, a set-valued observer, featuring a state predictor and a state estimator, is first constructed and designed to guarantee the inclusion of the unavailable actual battery state due to unknown modeling errors and noises at every instant of time. Compared with the traditional observers, a distinct feature of the proposed one lies in that the calculated state predictions and estimations of the battery system at each time step are ellipsoidal sets in state space rather than single vectors. The boundedness of state prediction and estimation errors is formally proved, and the tractable design criteria for determining the real-time optimal prediction and estimation ellipsoids are also derived. As for diagnosis algorithm, fault detection is implemented based on the intersection between the prediction and estimation ellipsoids. Then, a two-layer Pearson correlation coefficient analysis mechanism is developed to identify the source and type of sensor faults. Another set-valued observer based on an augmented battery model is further designed to estimate the fault level. Finally, experimental studies of a battery cell under different sensor fault sources, types and values are elaborated to verify the effectiveness of the proposed method.

**Index Terms**—Electric vehicle, fault detection, lithium-ion battery, sensor fault diagnosis, set-valued estimation, set-valued prediction, two-layer correlation coefficient analysis.

## NOMENCLATURE

|       |                                     |
|-------|-------------------------------------|
| EV    | electric vehicle.                   |
| LIB   | lithium-ion battery.                |
| SOC   | state of charge.                    |
| NEDC  | new European driving cycle.         |
| UDDS  | urban dynamometer driving schedule. |
| DST   | dynamic stress test.                |
| UBB   | unknown but bounded.                |
| $R_0$ | internal resistance.                |
| $R_p$ | polarization resistance.            |
| $C_p$ | polarization capacitance.           |
| $C_n$ | battery capacity.                   |

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|                   |                                    |
|-------------------|------------------------------------|
| $U_p$             | polarization voltage.              |
| $U_t$             | terminal voltage.                  |
| $U_{oc}$          | open circuit voltage.              |
| $I_L$             | internal current.                  |
| $x_k$             | true battery state.                |
| $\hat{x}_{k k}$   | estimated battery state.           |
| $\hat{x}_{k+1 k}$ | predicted battery state.           |
| $\omega_k$        | modeling error.                    |
| $d_k^i$           | process noise.                     |
| $d_{k+1}^v$       | measurement noise.                 |
| $x_k^{ag}$        | true augmented battery state.      |
| $\hat{x}_k^{ag}$  | estimated augmented battery state. |
| $\omega_k^{ag}$   | augmented modeling error.          |
| $d_k^{i,ag}$      | augmented process noise.           |

## I. INTRODUCTION

THE environmental crisis regarding greenhouse gas emissions has promoted the development of EVs [1], [2]. Due to high energy, power density and long lifespan, LIBs have become the dominant energy storage for EVs [3], [4]. To ensure the safe and reliable operation of LIBs, battery fault diagnosis has been investigated to avoid catastrophic damage to EVs [5], [6], [7]. It is clear that the sensitivity and accuracy of battery system fault diagnosis highly depend on the normal operation of current, voltage and temperature sensors [8], [9], [10]. It is common that these sensors experience certain malfunctions in realistic applications, which can cause mismanagement of battery cells (e.g. overcharge or overdischarge) and result in degraded battery performance or even serious safety hazards. An effective sensor fault diagnosis method is thus of great significance and necessity to improve battery operation safety [11].

Numerous fault diagnostic approaches for LIBs focus on battery faults [12], [14], [15], while few studies have been working on battery sensor faults [16]. Generally, the existing battery sensor fault diagnosis methods can be divided into two types [6]: model-free methods and model-based methods. As for model-free methods, Xia et al. [17], [18] devised an interleaved voltage measurement method for a series-connected battery pack, where the voltages of the multiple series-connected cells were measured and a voltage interpretation matrix was developed to detect faulty voltage sensors. Kang et al. [19] presented a diagnostic scheme by combining the cross-voltage measurement with the improved correlation coefficient analysis. The fault occurrence broke the battery consistency in a pack

which resulted in an abrupt drop in correlation coefficients. Accordingly, fault source and type were isolated and identified via comparisons between correlation coefficients and thresholds. Note that such a model-free method based on the consistency of battery electrical characteristics in a pack does not provide sensor fault estimation.

Regarding model-based methods, Xiong et al. [20] detected and identified current and voltage sensor faults in a series-connected battery pack based on the residual between SOC calculated by the Coulomb counting method and the SOC estimated by the unscented Kalman filter. Hu et al. [21] proposed a current sensor fault detection method by comparing the residuals with the fault-free range, where the residuals were generated from the estimated load currents and the measured load currents. Liu et al. [22] compared the residual between the estimated output voltage via an adaptive extended Kalman filter and the measured voltage with the pre-defined threshold to determine the presence of a fault and further identify fault source based on the analysis of the cause-effect relationship between fault and residuals. Tian et al. [23] predicted the thermal and electrical behaviors based on electrical and thermal models of a battery pack and compared the predicted terminal voltage, surface and core temperature with measurement to generate residual sets. According to the sensor fault isolation table, the fault source was then identified. Although the aforementioned methods can identify or detect sensor fault source based on the comparison of currents, voltages or temperatures, they are not directly applicable for sensor fault estimation. To cope with the fault estimation, Xu et al. [24] designed a proportional integral observer to simultaneously estimate current sensor bias fault and SOC for a faulty battery under the assumption of fault-free voltage sensors. The current sensor bias fault was then obtained from the integral of the voltage difference between the model and measurement. Dey et al. [25] presented a model-based sensor diagnosis method via sliding mode observers. According to the equivalent output error injections from observers, fault source and bias fault were identified and estimated. The above two methods are capable to provide sensor bias fault estimation. However, they neglect the inherent nonlinearity between SOC and battery system parameters, which may limit their practicability in EV applications and constrain their estimation accuracy in sensor fault severity evaluation.

Motivated by the observations aforementioned, in this paper, we address the sensor fault diagnosis problem for LIB packs in EVs. The novelty lies in a model-based sensor fault diagnosis method that empowers us to systematically detect the current and voltage sensor fault, identify the fault source (e.g., current or voltage sensor fault) and type (e.g., bias fault or scaling fault), and further estimate the fault level (i.e., fault severity) for a battery cell. The main technical contributions of this study are summarized as follows.

- 1) A general uncertain state-space model, incorporating the unknown modeling error as well as unknown process and measurement noises, is established to describe the electrical behavior of a battery. It will be shown that only the hard bounds of the unknown modeling errors and noises are required, while not demanding any *a priori*

knowledge of their statistical properties. This thus circumvents the stringent assumption on the accurate and linear time-invariant battery state-space models in the existing relevant literature.

- 2) A set-valued observer is designed to reconstruct the battery state. A salient feature of the designed predictor and estimator is that state prediction and state estimation at each time step are some ellipsoidal sets in state space, which are guaranteed to enclose the unavailable true battery system state and be immune to the effect of the unknown modeling error and noises. This is in contrast to the conventional state estimators or observers that only provide pointwise estimations in terms of single vectors at each time step.
- 3) A unified sensor fault diagnosis algorithm is developed to accomplish the simultaneous fault detection, fault identification, and fault estimation. Specifically, fault detection is implemented based on checking the intersection of two recursively calculated prediction and estimation ellipsoids, which is different from the existing residual comparison methods. A two-layer Pearson correlation coefficient analysis mechanism is then proposed for fault source and type identification. Once a sensor fault is detected and identified, another set-valued state observer is further constructed and designed to estimate fault value and level at each time step.
- 4) The efficacy and superiority of the proposed diagnosis method are validated via the experiments under multiple current profiles with various voltage and current sensor bias and scaling faults.

*Notations:* For a real vector  $c \in \mathbb{R}^n$  and a real-value matrix  $P = P^T > 0$ , an *ellipsoid* enclosing a real vector  $a \in \mathbb{R}^n$  is denoted by  $\xi(c, P, r) \triangleq \{a : (a - c)^T P^{-1} (a - c) \leq r\}$ , where  $P > 0$  represents the shape matrix,  $c$  represents the center,  $r > 0$  represents the radius of ellipsoid, respectively.

The remainder of this paper is organized as follows. The battery modeling is described in Section II. The set-valued observer design is presented in Section III. The sensor fault estimation result is discussed in Section IV. The sensor fault diagnosis algorithm is introduced in Section V. The experimental validation and discussion are provided in Section VI. The conclusion is drawn in Section VII.

## II. BATTERY MODELLING

A Thevenin battery model shown in Fig. 1 is adopted to describe the electrical dynamics of the battery. The model consists of an open circuit voltage source  $U_{oc}$  which has a nonlinear relationship with  $SOC$ , an internal resistance  $R_0$ , and a pair of polarization resistance and capacitance network ( $R_p, C_p$ ). Let  $i_L$  denote the internal current, which is positive for discharging and negative for charging,  $U_p$  and  $U_t$  represent the voltage across  $RC$  pair and terminal voltage, respectively. By virtue of Kirchhoff's law, one then has the following battery model

$$\begin{cases} \dot{U}_p = -\frac{1}{R_p C_p} U_p + \frac{1}{C_p} i_L, \\ U_t = U_{oc}(SOC) - U_p - R_0 i_L. \end{cases} \quad (1)$$

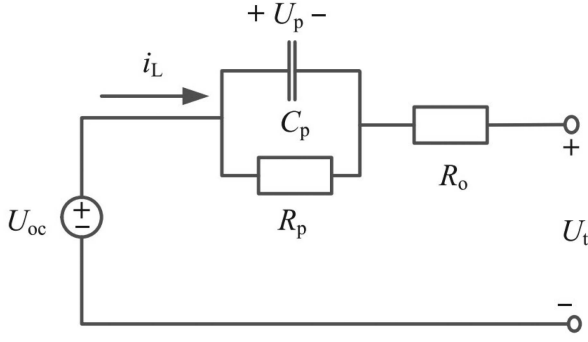


Fig. 1. Thevenin model of a lithium-ion battery.

According to the Coulomb counting method,  $\dot{SOC}$  can be expressed as  $\dot{SOC} = -\frac{i_L}{C_n}$ . Recalling (1), one can discretize the battery model with a sampling period of  $T_s$  seconds and further express it in a state-space form as [12]

$$x_{k+1} = A(x_{2,k})x_k + B(x_{2,k})u_k, \quad (2)$$

$$y_{k+1} = h(x_{2,k+1}) + Cx_{k+1} + D(x_{2,k+1})u_{k+1}, \quad (3)$$

where  $x_k = [x_{1,k}, x_{2,k}]^T = [U_p, SOC]^T$  denoting the true battery state,  $u_k = i_L$ ,  $h(x_{2,k+1}) = U_{oc}(SOC)$ , and the system matrices are described by

$$A(x_{2,k}) = \begin{bmatrix} e^{-\frac{T_s}{\tau_k}} & 0 \\ 0 & 1 \end{bmatrix}, \quad \tau_k = R_p(x_{2,k})C_p(x_{2,k}),$$

$$C = [-1, 0], \quad D(x_{2,k+1}) = -R_0(x_{2,k+1}),$$

$$B(x_{2,k}) = \left[ \left(1 - e^{-\frac{T_s}{\tau_k}}\right) R_p(x_{2,k}), -\frac{T_s}{C_n} \right]^T.$$

To deal with the nonlinear  $SOC$ -dependent parameters  $A(x_{2,k})$ ,  $B(x_{2,k})$ ,  $D(x_{2,k+1})$  and  $h(x_{2,k+1})$ , we apply Taylor Series expansion and linearize them with respect to  $x_{2,k}$  at  $\hat{x}_{2,k|k}$  and  $x_{2,k+1}$  at  $\hat{x}_{2,k+1|k}$ , where  $\hat{x}_{2,k|k}$  represents the estimated  $SOC$  at time  $k$  and  $\hat{x}_{2,k+1|k}$  denotes the predicted  $SOC$  at time  $k+1$  to be specified later. Considering the influences of modeling error and unknown noises, the uncertain battery model can be given as

$$x_{k+1} = A(\hat{x}_{2,k|k})x_k + B(\hat{x}_{2,k|k})u_k + \omega_k + B(\hat{x}_{2,k|k})d_k^i + \dot{B}(\hat{x}_{2,k|k})u_k(x_k - \hat{x}_{k|k}), \quad (4)$$

$$y_{k+1} = h(\hat{x}_{2,k+1|k}) + \dot{h}(\hat{x}_{2,k+1|k})(x_{k+1} - \hat{x}_{k+1|k}) + Cx_{k+1} + D(\hat{x}_{2,k+1|k})u_{k+1} + d_{k+1}^v + \dot{D}(\hat{x}_{2,k+1|k})u_{k+1}(x_{k+1} - \hat{x}_{k+1|k}), \quad (5)$$

where

$$\hat{x}_{k|k} = [\hat{x}_{1,k|k}, \hat{x}_{2,k|k}]^T, \quad \hat{x}_{k+1|k} = [\hat{x}_{1,k+1|k}, \hat{x}_{2,k+1|k}]^T,$$

$$\omega_k = A(x_{2,k})x_k - A(\hat{x}_{2,k|k})x_k,$$

$$\dot{B}(\hat{x}_{2,k|k}) = \begin{bmatrix} 0 & \frac{\partial B(x_{2,k})}{\partial x_{2,k}} \big|_{x_{2,k}=\hat{x}_{2,k|k}} \\ 0 & 0 \end{bmatrix},$$

$$\dot{h}(\hat{x}_{2,k+1|k}) = \left[ 0, \frac{\partial h(x_{2,k+1})}{\partial x_{2,k+1}} \big|_{x_{2,k+1}=\hat{x}_{2,k+1|k}} \right],$$

$$\dot{D}(\hat{x}_{2,k+1|k}) = \left[ 0, \frac{\partial D(x_{2,k+1})}{\partial x_{2,k+1}} \big|_{x_{2,k+1}=\hat{x}_{2,k+1|k}} \right].$$

Specifically, in (4) and (5),  $\omega_k$  stands for the unknown modeling error of the above linearization process,  $d_k^i$  denotes the unknown process noise affecting the battery state  $x_{k+1}$ , and  $d_{k+1}^v$  represents the unknown measurement noise corrupting  $y_{k+1}$ . It is assumed that these errors and noises are UBB and confined to the following ellipsoids

$$\omega_k \in \xi(0, Q_k, 1), \quad d_k^i \in \xi(0, O_k, 1), \quad d_{k+1}^v \in \xi(0, U_{k+1}, 1) \quad (6)$$

with  $Q_k$ ,  $O_k$  and  $U_{k+1}$  being the shape matrices for the ellipsoidal sets, respectively.

### III. SET-VALUED OBSERVER DESIGN

Due to the existence of the UBB modeling error  $\omega_k$ , process noise  $d_k^i$  and measurement noise  $d_{k+1}^v$ , the one-step ahead true battery system state  $x_{k+1}$  is inaccessible at each time step  $k+1$ . For this purpose, we develop a set-valued observer to reconstruct the battery state which can be further used for achieving fault diagnosis. Specifically, the set-valued observer consists of two parts: 1) a set-valued predictor; and 2) a set-valued estimator, which are given as follows.

*Set-Valued Predictor:*

$$\hat{x}_{k+1|k} = F_k \hat{x}_{k|k} + B(\hat{x}_{2,k|k})u_k, \quad (7)$$

$$\hat{y}_{k+1|k} = h(\hat{x}_{2,k+1|k}) + C\hat{x}_{k+1|k} + D(\hat{x}_{2,k+1|k})u_{k+1}; \quad (8)$$

*Set-Valued Estimator:*

$$\hat{x}_{k+1|k+1} = G_{k+1}\hat{x}_{k+1|k} + L_{k+1}(y_{k+1} - \hat{y}_{k+1|k}), \quad (9)$$

where  $\hat{y}_{k+1|k}$  denotes the predicted battery terminal voltage,  $\hat{x}_{k+1|k}$  denotes the prediction of the unavailable battery state  $x_{k+1}$  at time  $k+1$ ,  $\hat{x}_{k|k}$  denotes the estimation of the unavailable battery state  $x_k$  at time  $k$ ,  $\hat{x}_{k+1|k+1}$  denotes the estimate of the unavailable battery state  $x_{k+1}$  at time  $k+1$ , and  $F_k$  is the predictor gain to be designed, and  $G_{k+1}$  and  $L_{k+1}$  are the estimator gains to be designed.

Traditionally, the observer design aims to solve out the gain matrices  $F_k$ ,  $G_{k+1}$  and  $L_{k+1}$  such that the prediction  $\hat{x}_{k+1|k}$  and estimation  $\hat{x}_{k+1|k+1}$  can be determined at each time step  $k+1$ . In this case,  $\hat{x}_{k+1|k}$  and  $\hat{x}_{k+1|k+1}$  only represent a single prediction vector and a single estimation vector of the battery state  $x_{k+1}$  at time  $k+1$ . In this study, instead of determining *pointwise* predictions and estimations of the unavailable battery state at each time step, we aim to find the following two ellipsoids at each time step  $k+1$ .

$$\xi(\hat{x}_{k+1|k}, P_{k+1|k}, 1) \triangleq \left\{ x_{k+1} : (x_{k+1} - \hat{x}_{k+1|k})^T P_{k+1|k}^{-1} (x_{k+1} - \hat{x}_{k+1|k}) \leq 1 \right\},$$

$$\xi(\hat{x}_{k+1|k+1}, P_{k+1|k+1}, 1) \triangleq \left\{ x_{k+1} : (x_{k+1} - \hat{x}_{k+1|k+1})^T P_{k+1|k+1}^{-1} (x_{k+1} - \hat{x}_{k+1|k+1}) \leq 1 \right\},$$

where  $P_{k+1|k}$  and  $P_{k+1|k+1}$  denotes the real-valued positive shape matrices to be designed later. In this sense, at each time step  $k+1$ , the ellipsoidal prediction set  $\xi(\hat{x}_{k+1|k}, P_{k+1|k}, 1)$  and the ellipsoidal estimation set  $\xi(\hat{x}_{k+1|k+1}, P_{k+1|k+1}, 1)$  can provide 100% confidence to accommodate the unavailable battery state regardless of the UBB modeling error and noises [26]. This marks a distinct difference between the proposed set-valued observer and the existing pointwise observers.

To obtain the desired prediction and estimation ellipsoidal sets above, we make the following mild assumption.

**Assumption 1:** The unknown initial battery state  $x_0$  resides in the following ellipsoid  $\xi(\hat{x}_{0|0}, P_{0|0}, 1)$  with  $\hat{x}_{0|0}$  being the known center and  $P_{0|0}$  being a given shape matrix.

In what follows, we provided two design criteria for determining the above two ellipsoids. The gain parameters of the set-valued predictor (7) and set-valued estimator (9) also can be readily solved out from the two theorems below.

**Theorem 1:** Suppose that at time  $k$  the true battery system state  $x_k$  falls into the estimation ellipsoid  $\xi(\hat{x}_{k|k}, P_{k|k}, 1)$  with  $\hat{x}_{k|k}$  being the known center and  $P_{k|k}$  being the given shape matrix, if there exist matrices  $F_k, P_{k+1|k} > 0$  and scalars  $\theta_{m,k} > 0, m = 1, 2, 3$  such that the optimization problem

$$\begin{aligned} \min \text{trace}(P_{k+1|k}) \\ \text{s.t. } \Pi_{k+1|k} = \begin{bmatrix} -P_{k+1|k} & \tilde{\Pi}_k \\ * & \tilde{\Lambda}_k \end{bmatrix} < 0 \end{aligned} \quad (10)$$

is feasible, where

$$\begin{aligned} \tilde{\Pi}_k = [(A(\hat{x}_{2,k|k}) - F_k)\hat{x}_{k|k}, (A(\hat{x}_{2,k|k}) + \dot{B}(\hat{x}_{2,k|k})u_k)E_{k|k}, \\ I, B(\hat{x}_{2,k|k})], \end{aligned}$$

$$\tilde{\Lambda}_k = \text{diag} \left( \sum_{m=1}^3 \theta_{m,k} - 1, -\theta_{1,k}I, -\theta_{2,k}Q_k^{-1}, -\theta_{3,k}O_k^{-1} \right)$$

with  $E_{k|k}$  coming from the factorization  $P_{k|k} = E_{k|k}E_{k|k}^T$ , then the one-step ahead true battery state  $x_{k+1}$  is always enclosed by the optimal state prediction ellipsoid  $\xi(\hat{x}_{k+1|k}, P_{k+1|k}, 1)$  at time  $k+1$  under the UBB modeling error  $\omega_k$  and noise  $d_k^i$  defined in (6).

*Proof:* Recall that  $e_{k|k}^T P_{k|k}^{-1} e_{k|k} \leq 1$  at time  $k$ , where  $e_{k|k} = x_k - \hat{x}_{k|k}$ . By the Schur complement, one has that

$$e_{k|k} e_{k|k}^T \leq P_{k|k} = E_{k|k} E_{k|k}^T. \quad (11)$$

Setting  $\alpha_{1k} = E_{k|k}^{-1} e_{k|k}$ , we get that

$$\alpha_{1k}^T \alpha_{1k} = e_{k|k}^T E_{k|k}^{-T} E_{k|k}^{-1} e_{k|k} = e_{k|k}^T P_{k|k}^{-1} e_{k|k} \leq 1, \quad (12)$$

which means that there exists a vector  $\alpha_{1k}$  satisfying  $\|\alpha_{1k}\| \leq 1$  such that

$$x_k = \hat{x}_{k|k} + E_{k|k} \alpha_{1k}. \quad (13)$$

In light of the system model (4) and the state predictor (7), calculating the one-step ahead prediction error yields

$$\begin{aligned} e_{k+1|k} = x_{k+1} - \hat{x}_{k+1|k} = A(\hat{x}_{2,k|k})x_k - F_k \hat{x}_{k|k} + \omega_k \\ + B(\hat{x}_{2,k|k})d_k^i + \dot{B}(\hat{x}_{2,k|k})u_k E_{k|k} \alpha_{1k} \end{aligned}$$

$$= \tilde{\Pi}_k [1, \alpha_{1k}, \omega_k, d_k^i]^T \triangleq \tilde{\Pi}_k \phi_k. \quad (14)$$

Thus, the prediction error constraint  $e_{k+1|k}^T P_{k+1|k}^{-1} e_{k+1|k} \leq 1$  can be rewritten as

$$\phi_k^T (\tilde{\Pi}_k^T P_{k+1|k}^{-1} \tilde{\Pi}_k + \text{diag}(-1, 0, 0, 0)) \phi_k \leq 0. \quad (15)$$

On the other hand, from (6) and (12), one has that

$$\phi_k^T \text{diag}(1, 0, 0, -O_k^{-1}) \phi_k \geq 0, \quad (16)$$

$$\phi_k^T \text{diag}(1, 0, -Q_k^{-1}, 0) \phi_k \geq 0, \quad (17)$$

$$\phi_k^T \text{diag}(1, -I, 0, 0) \phi_k \geq 0. \quad (18)$$

Applying the  $\mathcal{S}$ -procedure, inequality (15) holds if there exist positive scalar sequences  $\theta_{m,k} > 0, m = 1, 2, 3$ , such that  $\phi_k^T \Delta_k \phi_k \leq 0$ , where

$$\begin{aligned} \Delta_k = \tilde{\Pi}_k^T P_{k+1|k}^{-1} \tilde{\Pi}_k + \text{diag}(-1, 0, 0, 0) \\ + \theta_{1,k} \text{diag}(1, -I, 0, 0) + \theta_{2,k} \text{diag}(1, 0, -Q_k^{-1}, 0) \\ + \theta_{3,k} \text{diag}(1, 0, 0, -O_k^{-1}). \end{aligned} \quad (19)$$

It is easy to see that  $\Delta_k \leq 0$  can be rewritten as (10) by using the Schur complement.

To guarantee an optimal state prediction ellipsoidal set, the trace of the shape matrix  $P_{k+1|k}$  is minimized at each time step  $k$ . This completes the proof. ■

**Theorem 2:** Suppose that at time  $k+1$  the true battery system state  $x_k$  resides in the state prediction ellipsoid  $\xi(\hat{x}_{k+1|k}, P_{k+1|k}, 1)$  with  $\hat{x}_{k+1|k}$  being the known center and  $P_{k+1|k}$  being the given shape matrix, if there exist matrices  $P_{k+1|k+1} > 0, G_{k+1}, L_{k+1}$ , and scalars  $\vartheta_{m,k} > 0, m = 1, 2$ , such that the optimization problem

$$\begin{aligned} \min \text{trace}(P_{k+1|k+1}) \\ \text{s.t. } \Pi_{k+1|k+1} = \begin{bmatrix} -P_{k+1|k+1} & \bar{\Pi}_{k+1} \\ * & \bar{\Lambda}_{k+1} \end{bmatrix} < 0 \end{aligned} \quad (20)$$

is feasible, where

$$\begin{aligned} \bar{\Pi}_{k+1} = [(I - G_{k+1})\hat{x}_{k+1|k}, (I - L_{k+1}(C + \dot{h}(\hat{x}_{2,k+1|k}) \\ + \dot{D}(\hat{x}_{2,k+1|k})u_{k+1}))E_{k+1|k}, -L_{k+1}], \end{aligned}$$

$$\bar{\Lambda}_{k+1} = \text{diag} \left( \sum_{m=1}^2 \vartheta_{m,k} - 1, -\vartheta_{1,k}I, -\vartheta_{2,k}U_{k+1}^{-1} \right)$$

with  $E_{k+1|k}$  coming from the factorization  $P_{k+1|k} = E_{k+1|k}E_{k+1|k}^T$ , then the one-step ahead true battery state  $x_{k+1}$  is also guaranteed to be enclosed by the optimal state estimation ellipsoid  $\xi(\hat{x}_{k+1|k+1}, P_{k+1|k+1}, 1)$  at time  $k+1$  in the presence of the UBB noise  $d_{k+1}^v$  defined in (6).

*Proof:* Recalling  $e_{k+1|k}^T P_{k+1|k}^{-1} e_{k+1|k} \leq 1$  at time  $k+1$ , one has that

$$e_{k+1|k} e_{k+1|k}^T \leq P_{k+1|k} = E_{k+1|k} E_{k+1|k}^T. \quad (21)$$

Therefore, there exists a vector  $\alpha_{2k}$  satisfying  $\|\alpha_{2k}\| \leq 1$  such that

$$x_{k+1} = \hat{x}_{k+1|k} + E_{k+1|k} \alpha_{2k}. \quad (22)$$



TABLE I  
 AUGMENTED BATTERY MODEL PARAMETERS

| Fault source | Fault type    | $c_k$                   | $a_k$                 |
|--------------|---------------|-------------------------|-----------------------|
| voltage      | bias fault    | 1                       | 0                     |
| voltage      | scaling fault | $y_k$                   | 0                     |
| current      | bias fault    | $-R_0(x_{2,k}^{ag})$    | $-\frac{T_s}{C_n}$    |
| current      | scaling fault | $-R_0(x_{2,k}^{ag})u_k$ | $-\frac{T_s}{C_n}u_k$ |

Calculating the one-step ahead state estimation error yields that

$$\begin{aligned} e_{k+1|k+1} &= x_{k+1} - \hat{x}_{k+1|k+1} \\ &= \hat{x}_{k+1|k} + E_{k+1|k}\alpha_2 k - F_k \hat{x}_{k+1|k} \\ &\quad - L_k(y_{k+1} - \hat{y}_{k+1|k}) \triangleq \bar{\Pi}_{k+1}\phi_{k+1}, \end{aligned} \quad (23)$$

where  $\phi_{k+1} = [1, \alpha_2 k, d_{k+1}^v]^T$ . The rest of the proof follows the similar procedure as that of Theorem 1, which leads to (20). This completes the proof. ■

*Remark 1:* It is clear from Theorems 1 and 2 that the concerned set-valued observer design problem of the battery system is tactfully cast into the optimization problems of a set of recursive linear matrix inequalities in (10) and (20), through which the predictor and estimator gain matrices and the shape matrices for the optimal ellipsoids can be determined recursively. It is also shown that the proposed set-valued ellipsoidal observer design method can deal with the inherently unknown yet bounded modeling error and (process and measurement) noises. Their adverse effects on the state prediction and estimation performance of the battery system can be suitably resisted since the design criteria above incorporate the error and noise bound information. Hence, our observer design method for LIBs enjoys certain resilience and robustness against UBB modeling error and noises [27]. Moreover, the designed state predictor and estimator eliminate the stringent requirement of an accurate and linear battery system model that has been universally assumed in the literature. Another prominent feature of the proposed observer design method lies in its confidence and guarantee to enclose the unknown true battery system state  $x_{k+1}$  at each time step  $k+1$  in the presence of the UBB modeling error and noises. This is because the calculated state predictions and estimations are ellipsoidal sets in state space rather than single vectors determined by some traditional estimation methods such as  $H_\infty$  observers [28], [29] and Kalman filters [30], [31].

#### IV. SENSOR FAULT ESTIMATION

Considering the battery suffering in a sensor fault  $f_k$ , the battery model (2) and (3) becomes

$$x_{k+1} = A(x_{2,k})x_k + B(x_{2,k})u_k + R_k f_k, \quad (24)$$

$$y_k = h(x_{2,k}) + Cx_k + D(x_{2,k})u_k + c_k f_k, \quad (25)$$

where  $R_k = [0, a_k]^T$  and  $a_k, c_k$  are determined by the sensor fault types which are clarified in Table I. To estimate the unknown sensor fault  $f_k$ , we rewrite the above state-space model

into an augmented form as

$$x_{k+1}^{ag} = \mathbf{A}(x_{2,k}^{ag})x_k^{ag} + \mathbf{B}(x_{2,k}^{ag})u_k, \quad (26)$$

$$y_k = h(x_{2,k}^{ag}) + \mathbf{C}_k x_k^{ag} + D(x_{2,k}^{ag})u_k \quad (27)$$

under the fault dynamics  $f_{k+1} = f_k$ , where  $x_k^{ag} = [x_{1,k}^{ag}, x_{2,k}^{ag}, x_{3,k}^{ag}]^T = [U_p, SOC, f_k]^T$  denotes the augmented battery state, and  $h(x_{2,k}^{ag}) = U_{oc}(SOC)$ . The system matrices are described as

$$\mathbf{A}(x_{2,k}^{ag}) = \begin{bmatrix} e^{-\frac{T_s}{\tau_k}} & 0 & 0 \\ 0 & 1 & a_k \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{C}_k = [-1, 0, c_k],$$

$$\mathbf{B}(x_{2,k}^{ag}) = \left[ \left(1 - e^{-\frac{T_s}{\tau_k}}\right) R_p(x_{2,k}^{ag}), -\frac{T_s}{C_n}, 0 \right]^T,$$

$$D(x_{2,k}^{ag}) = -R_0(x_{2,k}^{ag}), \quad \tau_k = R_p(x_{2,k}^{ag}) C_p(x_{2,k}^{ag}).$$

Similar to (4) and (5), after linearization at  $\hat{x}_k^{ag} = [\hat{x}_{1,k}^{ag}, \hat{x}_{2,k}^{ag}, \hat{x}_{3,k}^{ag}]^T$  and considering the influence of modeling error and noises, the state-space formulation of the augmented battery model (26) can be given as

$$\begin{aligned} x_{k+1}^{ag} &= \mathbf{A}(\hat{x}_{2,k}^{ag})x_k^{ag} + \mathbf{B}(\hat{x}_{2,k}^{ag})u_k + \omega_k^{ag} \\ &\quad + \mathbf{B}(\hat{x}_{2,k}^{ag})d_k^{i,ag} + \dot{\mathbf{B}}(\hat{x}_{2,k}^{ag})u_k e_k^{ag}, \end{aligned} \quad (28)$$

$$\begin{aligned} y_k &= h(\hat{x}_{2,k}^{ag}) + \mathbf{C}_k x_k^{ag} + D(\hat{x}_{2,k}^{ag})u_k + d_k^v \\ &\quad + \dot{\mathbf{D}}(\hat{x}_{2,k}^{ag})u_k e_k^{ag} + \dot{\mathbf{h}}(\hat{x}_{2,k}^{ag})e_k^{ag}, \end{aligned} \quad (29)$$

where

$$\dot{\mathbf{B}}(\hat{x}_{2,k}^{ag}) = \begin{bmatrix} 0 & \frac{\partial B(x_{2,k}^{ag})}{\partial (x_{2,k}^{ag})} \Big|_{x_{2,k}^{ag}=\hat{x}_{2,k}^{ag}} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

$$\dot{\mathbf{D}}(\hat{x}_{2,k}^{ag}) = \begin{bmatrix} 0 \\ \frac{\partial D(x_{2,k}^{ag})}{\partial (x_{2,k}^{ag})} \Big|_{x_{2,k}^{ag}=\hat{x}_{2,k}^{ag}} \\ 0 \end{bmatrix},$$

$$\dot{\mathbf{h}}(\hat{x}_{2,k}^{ag}) = \begin{bmatrix} 0 \\ \frac{\partial h(x_{2,k}^{ag})}{\partial (x_{2,k}^{ag})} \Big|_{x_{2,k}^{ag}=\hat{x}_{2,k}^{ag}} \\ 0 \end{bmatrix},$$

$$\omega_k^{ag} = \mathbf{A}(x_{2,k}^{ag})x_k^{ag} - \mathbf{A}(\hat{x}_{2,k}^{ag})x_k^{ag} \in \xi(0, Q_k^{ag}, 1),$$

$$e_k^{ag} = x_k^{ag} - \hat{x}_k^{ag}, \quad d_k^{i,ag} \in \xi(0, O_k^{ag}, 1),$$

$$d_k^v \in \xi(0, U_k, 1).$$

To estimate the augmented state  $x_{k+1}^{ag}$ , the following set-valued observer is constructed

$$\hat{x}_{k+1}^{ag} = \mathbf{A}(\hat{x}_{2,k}^{ag})\hat{x}_k^{ag} + \mathbf{B}(\hat{x}_{2,k}^{ag})u_k + \mathbf{L}_{k+1}(y_k - \hat{y}_k), \quad (30)$$

$$\hat{y}_k = h(\hat{x}_{2,k}^{ag}) + \mathbf{C}_k \hat{x}_k^{ag} + D(\hat{x}_{2,k}^{ag})u_k, \quad (31)$$

where  $\mathbf{L}_{k+1}$  is the observer gain to be designed. The aim is to determine an ellipsoidal estimation set at each time step  $k + 1$

$$\xi(\hat{x}_{k+1}, P_{k+1}, 1) \triangleq \{x_{k+1} : (e_{k+1}^{ag})^T P_{k+1}^{-1} e_{k+1}^{ag} \leq 1\},$$

where  $P_{k+1}$  is the real-valued positive shape matrix to be specified later.

We next present the following theorem which outlines a sufficient condition on the existence of the optimal state estimation ellipsoid  $\xi(\hat{x}_{k+1}, P_{k+1}, 1)$  and the desired observer in the form of (30) and (31).

**Theorem 3:** Suppose that at time  $k$  the true battery system state  $x_k$  belongs to the known estimation ellipsoid  $\xi(\hat{x}_k, P_k, 1)$  with  $\hat{x}_k$ , if there exist matrices  $\mathbf{L}_{k+1}$ ,  $P_{k+1} > 0$  and scalars  $\kappa_{m,k} > 0$ ,  $m = 1, 2, 3, 4$  such that the optimization problem

$$\begin{aligned} \min \text{trace}(P_{k+1}) \\ \text{s.t. } \Pi_{k+1} = \begin{bmatrix} -P_{k+1} & \hat{\Pi}_{k+1} \\ * & \hat{\Lambda}_{k+1} \end{bmatrix} < 0 \end{aligned} \quad (32)$$

is feasible, where

$$\begin{aligned} \hat{\Pi}_{k+1} &= \begin{bmatrix} 0, \left( \mathbf{A}(\hat{x}_{2,k}^{ag}) + \dot{\mathbf{B}}(\hat{x}_{2,k}^{ag}) u_k - \mathbf{L}_{k+1} (\mathbf{C}_k + \dot{\mathbf{h}}(\hat{x}_{2,k}^{ag}) \right. \\ \left. + \dot{\mathbf{D}}(\hat{x}_{2,k}^{ag}) u_k) \right) E_k, I, \mathbf{B}(\hat{x}_{2,k}^{ag}), -\mathbf{L}_{k+1} \end{bmatrix}, \\ \hat{\Lambda}_{k+1} &= \text{diag} \left( \sum_{m=1}^4 \kappa_{m,k} - 1, -\kappa_{1,k} I, -\kappa_{2,k} (Q_k^{ag})^{-1}, \right. \\ &\quad \left. -\kappa_{3,k} (O_k^{ag})^{-1}, -\kappa_{4,k} U_k^{-1} \right) \end{aligned}$$

with  $E_k$  coming from the factorization  $P_k = E_k E_k^T$ , then the one-step ahead true battery state  $x_{k+1}$  is always enclosed by the optimal state estimation ellipsoid  $\xi(\hat{x}_{k+1}, P_{k+1}, 1)$  at time  $k + 1$  even under the UBB modeling error  $\omega_k^{ag}$  and noises  $d_k^{i,ag}$  and  $d_k^v$ .

**Proof:** Recalling  $e_k^T P_k^{-1} e_k \leq 1$  at time  $k$ , one has that  $e_k^{ag} (e_k^{ag})^T \leq P_k = E_k E_k^T$ . Then, there exists a vector  $\alpha_3 k$  satisfying  $\|\alpha_3 k\| \leq 1$  such that  $x_k^{ag} = \hat{x}_k^{ag} + E_k \alpha_3 k$ . Calculating the one-step ahead estimate error yields that

$$\begin{aligned} e_{k+1}^{ag} &= \mathbf{A}(\hat{x}_{2,k}^{ag}) E_{k+1} \alpha_3 k + \omega_k^{ag} - \mathbf{L}_{k+1} (y_k - \hat{y}_k) \\ &\quad + \mathbf{B}(\hat{x}_{2,k}^{ag}) d_k^{i,ag} + \dot{\mathbf{B}}(\hat{x}_{2,k}^{ag}) u_k e_k^{ag} \triangleq \hat{\Pi}_{k+1} \phi_{k+1}, \end{aligned} \quad (33)$$

where  $\phi_{k+1} = [1, \alpha_3 k, \omega_k^{ag}, d_k^{i,ag}, d_k^v]^T$ . The rest of the proof follows the similar procedure as that of Theorem 1, which leads to (32). This completes the proof. ■

## V. A SENSOR FAULT DIAGNOSIS ALGORITHM

In this section, we develop a sensor fault diagnosis algorithm, namely Algorithm 1, which computes the gain matrices of the desired set-valued observers for state prediction and estimation purposes. It is further shown that the algorithm allows us to achieve comprehensive fault diagnosis, including fault

### Algorithm 1: Battery Sensor Fault Diagnosis.

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**Input:** current profile  $i_L$ , polynomial parameters  $R_0, R_p, C_p, U_{oc}$ , maximal simulation time  $T_{max}$ , moving window size  $W_{in}$ , initial parameters  $\hat{x}_{0|0}, SOC_{ref,0}, Th_1, k_f = 0, T_s = 1$

**Output:**  $\hat{x}_{2,k}$  (SOC);  $\hat{x}_{2,k}^{ag}$  (SOC);  $\hat{x}_{3,k}^{ag}$  (sensor fault)

---

```

1  \ % Sensor fault detection %\
2  for  $k = 0 : T_{max}$  do
3    \ % State prediction %\
4    apply  $u_k = i_L(k)$ ;
5    update  $A(\hat{x}_{2,k|k}), B(\hat{x}_{2,k|k}), D(\hat{x}_{2,k|k}), \dot{B}(\hat{x}_{2,k|k})$ ;
6    solve Theorem 1 to get  $F_k$ ;
7    calculate  $\hat{x}_{k+1|k}$  and  $\hat{y}_{k+1|k}$  by (7) and (8);
8    \ % State estimation %\
9    apply  $u_{k+1} = i_L(k+1)$ ;
10   measure the terminal voltage  $y_{k+1}$  and update model
       parameters  $\dot{D}(\hat{x}_{2,k+1|k})$  and  $\dot{h}(\hat{x}_{2,k+1|k})$ ;
11   solve Theorem 2 to get  $G_{k+1}, L_{k+1}$ ;
12   calculate  $\hat{x}_{k+1|k+1}$  by (9);
13   if  $|\hat{x}_{2,k+1|k} - \hat{x}_{2,k+1|k+1}| \geq Th_1$  then
14     display "Battery in Faulty State";
15     record fault duration  $k_f = k_f + 1$ ;
16     compute  $SOC_{ref,k+1} = SOC_{ref,k} - u_k/C_n$ ;
17     compute  $\dot{SOC}_{k+1} = \hat{x}_{2,k+1|k+1} - SOC_{ref,k+1}$ ;
18   else
19     display "Battery in Healthy State";
20   end
21 end
22 if  $k_f == 0$  then
23   Exit;
24 else
25   \ % Fault source and type identification %\
26   \ % First-layer correlation coefficient analysis %\
27   for  $k = (T_{max} - k_f + 1) : (T_{max} - W_{in})$  do
28     calculate correlation coefficient  $(r_{\dot{SOC},k})_k$ ;
29   end
30   \ % Second-layer correlation coefficient analysis %\
31   for  $k = (T_{max} - k_f + 1) : (T_{max} - W_{in})$  do
32     calculate correlation coefficient  $(r_{\dot{SOC},y_k})_k$ ;
33   end
34   identify the fault type according to the correlation coefficients;
35   build the augmented battery model  $\hat{x}_k^{ag} = (U_p, SOC, f_k)$ ;
36   \ % Sensor fault estimation %\
37   initialize  $\hat{x}_{2,T_{max}-k_f+2}^{ag} = \hat{x}_{2,T_{max}-k_f+2|T_{max}-k_f+2}$ ;
38   for  $k = (T_{max} - k_f + 2) : T_{max}$  do
39     apply  $u_k = i_L(k)$ ;
40     update model parameters in (28) and (29);
41     calculate  $\hat{y}_k$  by (31);
42     solve Theorem 3 to get  $\mathbf{L}_{k+1}$ ;
43     calculate  $\hat{x}_{k+1}^{ag}$  by (30);
44   end
45 end

```

---

detection, fault source and type identification, and sensor fault estimation.

The rationale of the sensor fault detection mechanism in Algorithm 1 lies in the verification of the center bias of the prediction and estimation ellipsoids with some pre-defined threshold  $Th_1$ . This is because the center  $\hat{x}_{k+1|k+1}$  of the estimation ellipsoid may be abruptly altered due to the sensor fault at time  $k + 1$ , while the center  $\hat{x}_{k+1|k}$  of the prediction ellipsoid remains in a healthy condition as the previous state estimation  $\hat{x}_{k|k}$  is normal. As a result, two ellipsoids may have a sufficient center bias or even an empty intersection at time  $k + 1$ . In this sense, sensor fault occurrence can be effectively detected via verifying the detection criterion  $|\hat{x}_{2,k+1|k} - \hat{x}_{2,k+1|k+1}| \geq Th_1$ .

To further identify fault source and type, a two-layer correlation coefficient analysis mechanism is devised in Algorithm 1. Specifically, the correlation coefficient analysis mechanism is

TABLE II  
 SPECIFICATION OF A LiCoO<sub>2</sub> BATTERY

| Items                       | Specification |
|-----------------------------|---------------|
| Nominal capacity            | 2.2 Ah        |
| Nominal voltage             | 3.6 V         |
| Operating temperature range | -20°C - 60°C  |
| Maximum operating range     | 3.0 V - 4.2 V |
| Maximum discharge rate      | 2C            |

 TABLE III  
 DESCRIPTION OF EXPERIMENTAL CONDITIONS

|        | Fault source   | Fault type    | Fault value |
|--------|----------------|---------------|-------------|
| Case 1 | voltage sensor | bias fault    | -0.1V       |
| Case 2 | voltage sensor | bias fault    | -0.2V       |
| Case 3 | current sensor | bias fault    | -0.2A       |
| Case 4 | current sensor | bias fault    | -0.5A       |
| Case 5 | voltage sensor | scaling fault | -10%        |
| Case 6 | voltage sensor | scaling fault | -20%        |
| Case 7 | current sensor | scaling fault | -10%        |
| Case 8 | current sensor | scaling fault | -20%        |

firstly applied on SOC difference and sampling time index  $r\hat{SOC}_{k,k}$ . The current bias fault and voltage scaling fault can be directly identified after the first-layer analysis, while the voltage bias fault and current scaling fault generally share similar responses. It thus requires further analysis, namely the second-layer analysis, between SOC difference and terminal voltage  $r\hat{SOC}_{k,y_k}$  to accomplish successful identification. Under such a two-layer correlation coefficient analysis, four types of sensor faults, including voltage sensor bias and scaling, current sensor bias and scaling, can be distinguished effectively.

As the last step of the diagnosis algorithm, an augmented set-valued observer of the form (30) and (31) is designed to estimate simultaneously the sensor fault and battery state after sensor fault type and source identification step.

## VI. EXPERIMENTAL VALIDATION AND DISCUSSION

To illustrate the effectiveness of the proposed battery sensor fault diagnosis algorithm, experiments are conducted under three EV current profiles based on NEDC, DST and UDDS. The experimental setup consists of three parts [12], [13], including 1) a workbench, which provides an insulated working environment; 2) a battery test system (Arbin BT2000 20V-100A@4CH), which programs the current profile for battery with the accuracy up to 0.02 %; 3) a host computer, which controls the current coming from the battery test system. It is found that the proposed algorithm demonstrates similar diagnosis performance under NEDC, DST and UDDS, thus NEDC is considered as a representative example to demonstrate the performance of the designed algorithm in EVs.

Table II shows the specification of 18650 type LiCoO<sub>2</sub> battery applied in this study. The experimental conditions including fault source, fault type and fault value are summarized in Table III. For voltage sensor faults, as the normal operating voltage range for this battery is from 3.58 V to 4 V corresponding to the SOC region from 20% to 80%, we select 0.2 V as the upper limit voltage bias to ensure that the battery will not suffer from overcharge. For current sensor faults, we make an analogy

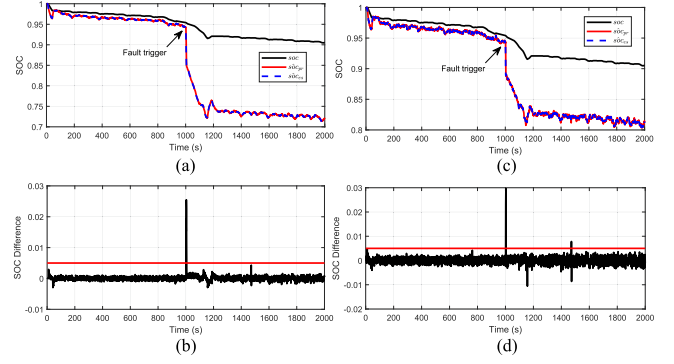


Fig. 2. Voltage sensor bias fault detection under NEDC: (a) SOC in case 1; (b) SOC difference in case 1; (c) SOC in case 2; (d) SOC difference in case 2.

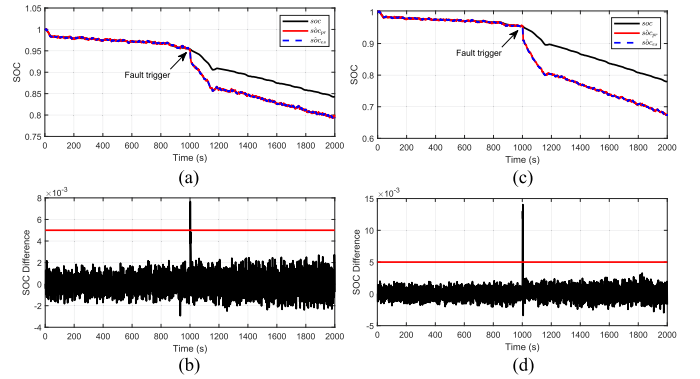


Fig. 3. Current sensor bias fault detection under NEDC: (a) SOC in case 3; (b) SOC difference in case 3; (c) SOC in case 4; (d) SOC difference in case 4.

between a current bias fault and a short circuit fault. The large bias fault current is similar to a hard short circuit fault. Since the nominal voltage for the battery is about 3.6 V and 8 Ω represents a critical value differentiating hard short circuit from soft short circuit [32], the current sensor bias fault is set as 0.5 A (3.6 V/8 Ω ≈ 0.5 A). In this paper, we also choose 0.1 V and 0.2 A to show that the proposed method can be applied for sensor fault diagnosis at different levels of severity. All the sensor fault tests are performed at 1000 seconds and last until the terminal voltage reaches the cut-off value.

Then, we let  $d_k^i$  be a random current disturbance fluctuating between  $(-0.01 \text{ A}, 0.01 \text{ A})$  and  $d_{k+1}^v$  be a random voltage disturbance fluctuating between  $(-0.01 \text{ V}, 0.01 \text{ V})$ . The matrices are chosen as  $Q_k = O_k = \text{diag}(0.1, 0.1)$ ,  $U_{k+1} = 0.1$ , and  $Q_k^{ag} = O_k^{ag} = \text{diag}(0.1, 0.1, 0.0001)$ . To achieve a well balance between a false alarm and a missing alarm, we select the fault detection threshold  $Th_1$  as  $5 \times 10^{-3}$  by trial and error. The window size  $W_{in}$  for correlation coefficient calculation is selected to be 100 seconds.

For fault detection in Case 1 and Case 2, the SOC trajectories provided by the set-valued predictor and estimator together with the actual SOC are plotted in the subplots (a) and (c) in Figs. 2, 3, 4, and 5, where actual SOC are denoted by the black line, SOC given by the predictor ( $\hat{SOC}_{pr}$ ) and estimator ( $\hat{SOC}_{es}$ ) are respectively represented in the red and blue dash line. Their corresponding SOC differences are shown in the subplots (b) and

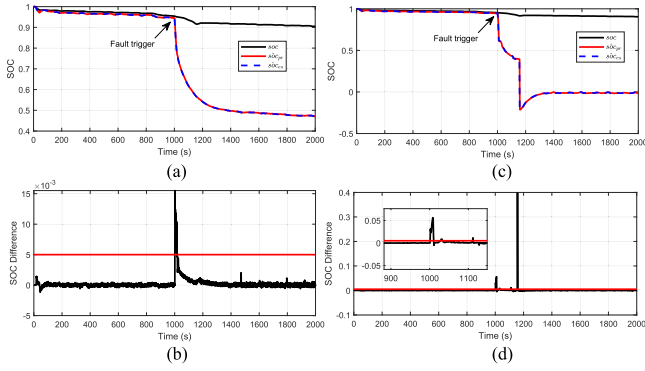


Fig. 4. Voltage sensor scaling fault detection under NEDC: (a) SOC in case 5; (b) SOC difference in case 5; (c) SOC in case 6; (d) SOC difference in case 6.

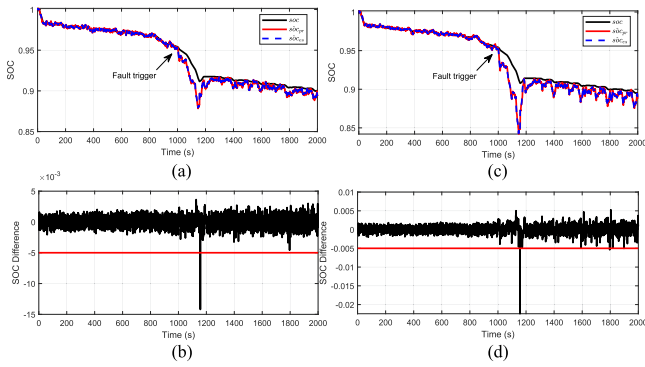


Fig. 5. Current sensor scaling fault detection under NEDC: (a) SOC in case 7; (b) SOC difference in case 7; (c) SOC in case 8; (d) SOC difference in case 8.

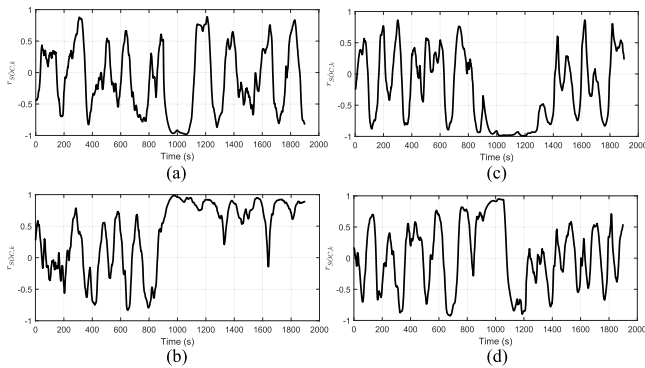


Fig. 6. Correlation coefficient analysis between SOC difference and the sampling time index: (a) voltage bias fault of  $-0.1$  V; (b) voltage scaling fault 10% off scaling; (c) current bias fault of  $-0.2$  A; (d) current scaling fault 10% off scaling.

(d) in Figs. 2–5. It can be seen that at the time of fault occurrence, the SOC difference between the prediction and estimation has an abrupt change which is far more than the threshold  $5 \times 10^{-3}$ , indicating the detection of a sensor fault.

Fig. 6(a)–(d) show the first-layer correlation coefficient analysis between the SOC difference and sampling time index for the voltage sensor with the bias fault of  $-0.1$  V, the voltage sensor with 10% off scaling fault, the current sensor with the bias fault of  $-0.2$  A and the current sensor with 10% off scaling fault, respectively. Fig. 7(a)–(d) are correlation coefficients between

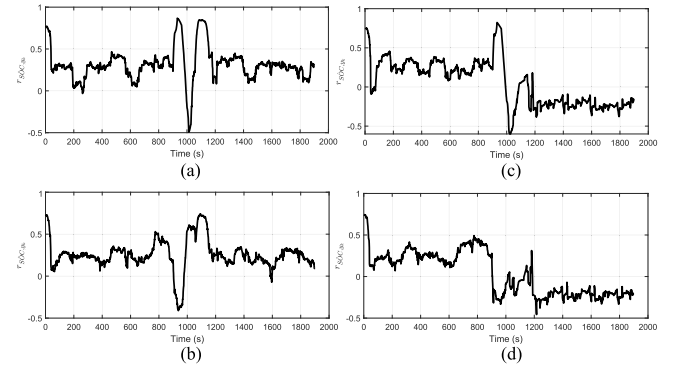


Fig. 7. Correlation coefficient analysis between SOC difference and terminal voltage: (a) voltage bias fault of  $-0.1$  V; (b) voltage scaling fault 10% off scaling; (c) current bias fault of  $-0.2$  A; (d) current scaling fault 10% off scaling.

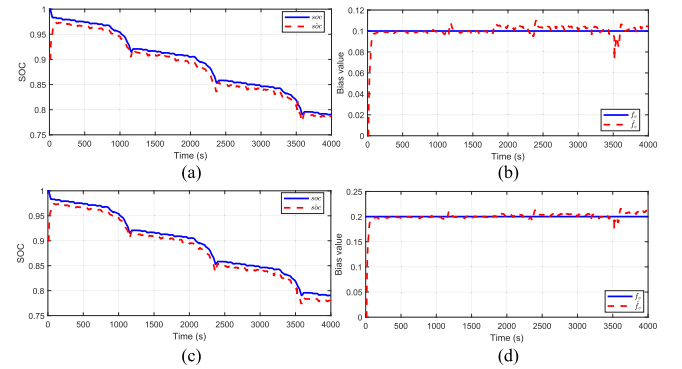


Fig. 8. Voltage sensor bias fault value estimation: (a) SOC under faulty condition in case 1; (b) voltage sensor bias fault estimation in case 1; (c) SOC under faulty condition in case 2; (d) voltage sensor bias fault estimation in case 2.

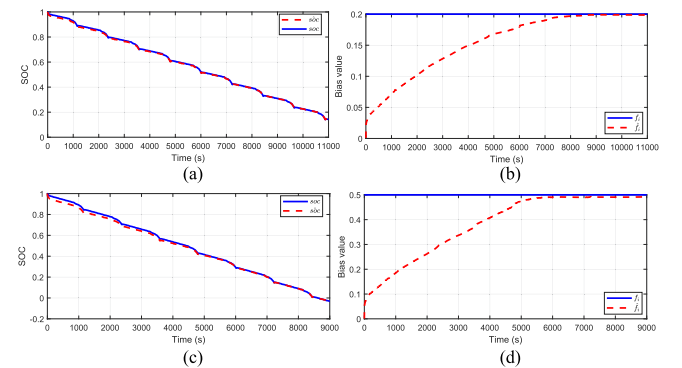


Fig. 9. Current sensor bias fault value estimation: (a) SOC under faulty condition in case 3; (b) current sensor bias fault estimation in case 3; (c) SOC under faulty condition in case 4; (d) current sensor bias fault estimation in case 4.

the SOC difference and terminal voltage. According to the transient and steady state responses of the correlation coefficient from the first-layer and from the second-layer, fault source and fault type can be identified successfully.

The sensor fault estimation results in Cases 1–8 are shown in Figs. 8, 9, 10, and 11. It can be found that the error for SOC estimation under current sensor faults is within 1%, while under voltage sensor faults is about 5%, especially for the voltage



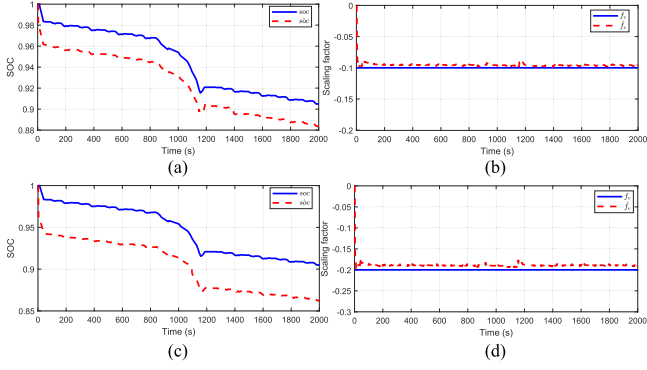


Fig. 10. Voltage sensor scaling fault value estimation: (a) SOC under faulty condition in case 5; (b) scaling factor estimation in case 5; (c) SOC under faulty condition in case 6; (d) scaling factor estimation in case 6.

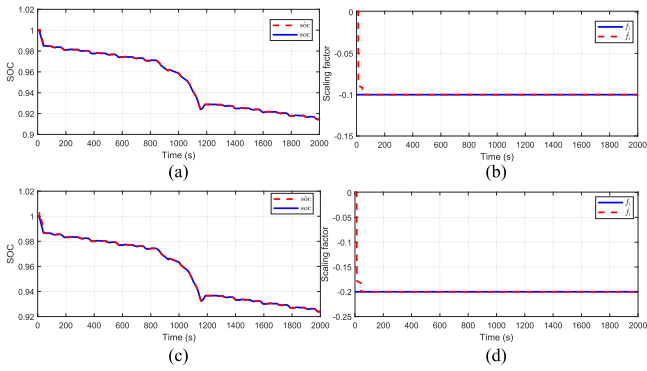


Fig. 11. Current sensor scaling fault value estimation: (a) SOC under faulty condition in case 7; (b) scaling factor estimation in case 7; (c) SOC under faulty condition in case 8; (d) scaling factor estimation in case 8.

scaling fault. Moreover, the accuracy of sensor fault estimation is almost kept above 95 %. Thus, accurate estimation for sensor fault and SOC under faulty conditions in all the cases shown in Figs. 8–11 verifies the effectiveness of the proposed set-valued prediction and estimation-based fault diagnosis algorithm.

Regarding the fault detection in terms of voltage sensor faults in Cases 1–2 and 5–6, it can be seen that the value of  $|\hat{SOC}_{pr} - \hat{SOC}_{es}|$  caused by current sensor fault in Cases 3–4 and 7–8 is relatively imperceptible. This is due to the following two reasons: 1) in the designed observer, a correction term of the terminal voltage  $y_{k+1}$  comes from voltage measurement, the normal voltage measurement maintains high SOC estimation even under current sensor faulty condition; and 2) when mA current fault is used to calculate SOC, the influence on SOC estimation is very small due to the factor of  $-1/C_n$ . Whereas, for mV voltage fault, the corresponding SOC variance becomes more appreciable from interpretation of the OCV-SOC curve. Therefore, the proposed fault detection method has a higher sensitivity towards voltage sensor faults than current sensor faults.

In order to evaluate different correlation coefficient responses under various sensor faults simulated above, we define the transient state for the first 200 seconds after the fault occurrence and the steady state for the last 1200 seconds to the end of the experiments. It can be observed that once faults are triggered

TABLE IV  
PERFORMANCE COMPARISON UNDER EV CURRENT PROFILES

| Current profiles | Fault type            | Mean estimation error |
|------------------|-----------------------|-----------------------|
| UDDS             | current sensor (0.5A) | 2.4%                  |
|                  | current sensor (10%)  | 0.7%                  |
|                  | voltage sensor (0.2V) | 4.3%                  |
|                  | voltage sensor (10%)  | 1.0%                  |
| NEDC             | current sensor (0.5A) | 1.7%                  |
|                  | current sensor (10%)  | 0.9%                  |
|                  | voltage sensor (0.2V) | 2.0%                  |
|                  | voltage sensor (10%)  | 3.9%                  |
| DST              | current sensor (0.5A) | 1.6%                  |
|                  | current sensor (10%)  | 1.2%                  |
|                  | voltage sensor (0.2V) | 10.5%                 |
|                  | voltage sensor (10%)  | 3.8%                  |

at 1000 seconds, their correlation coefficient responses can be examined to identify the fault source and fault type. Specifically, the current bias fault in Cases 3–4 makes the SOC difference proportional to the sampling time so that the correlation coefficients fluctuate along one in the steady state as shown in Fig. 6(c). This indicates the strong correlation between SOC difference and sampling time index. For voltage scaling fault in Cases 5–6, the correlation coefficients approach to  $-1$  during the transient state as shown in Fig. 6(b). Thus, the above two types of faults can be identified by the first layer of the correlation coefficient analysis. On the other hand, for the voltage bias fault and current scaling fault, their correlation coefficients fluctuate between 1 and  $-1$  disorderedly as shown in Fig. 6(a) and 6(d). Therefore, the second layer correlation coefficient analysis between the SOC differences and terminal voltages is required. As shown in Fig. 7(a), when the voltage bias fault in Cases 1–2 occurs, the correlation coefficients have a drastic spike in the transient state. Additionally, the estimated SOC under the voltage sensor bias fault has a constant difference with the calculated SOC, as shown in Fig. 2(a) and 2(c). In this sense, the correlation coefficients under voltage bias faults recover to the original shape in the steady state. Since then, the remaining fault is the current scaling fault in Cases 7–8.

Finally, as for the state estimation performance, it can be observed from Figs. 8–11(a) and 11(c) that the SOC estimations under current sensor faults in Cases 2–3 and 7–8 exhibit a higher accuracy than those under voltage sensor faults in Cases 1–2 and 5–6, due to the correction coming from the terminal voltage. Together with the detection sensitivity aforementioned, it can be concluded that the proposed method guarantees higher sensitivity to voltage sensor faults than current sensor faults, and more accurate state estimations under current sensor faults than under voltage sensor faults.

The experiments are also repeated under the EV current profile based on UDDS. For brevity, we only show the comparison of the mean estimation errors for four different types of sensor fault under three EV current profiles in Table IV, which indicates that the proposed algorithm demonstrates satisfactory and consistent performance.

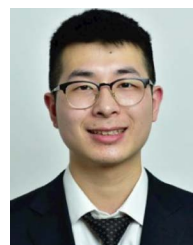
## VII. CONCLUSION

A novel sensor fault diagnosis algorithm is proposed to detect, identify and estimate battery sensor faults for LIB packs in EVs. Specifically, a set-valued detection mechanism, featuring

a set-valued predictor and a set-valued estimator, is devised to determine the occurrence of any sensor fault. Once a sensor fault is detected, a two-layer Pearson correlation coefficient analysis mechanism is then applied to identify the fault source and type. After that, a set-valued observer based on an augmented battery state is constructed and designed to simultaneously estimate the sensor fault and SOC under faulty conditions. The effectiveness of the proposed sensor fault diagnosis algorithm is verified by the extensive experimental case studies of an EV driving profile. The experimental results demonstrate that the proposed diagnosis method guarantees fast sensor fault detection within 5 seconds regardless of the sensor fault source, fault type and fault severity. Moreover, it can accurately estimate the sensor fault with a mean error less than 4.3% and estimate the SOC under faulty conditions with an accuracy higher than 95%.

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